INTERNSHIP REPORT ON

MICROSOST AZURE AI INTERN

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Duration: 10.04.2025 to 15.05.2025

College Name: Loyola-ICAM College of Engineering and Technology, Chennai.

Company Name: Edunet Foundations

Acknowledgement

I would like to express my sincere gratitude to Edunet Foundations for providing me with the opportunity to do internship as an AI Intern .I am also deeply thankful to my project guide, Mr. Rohit Shelar, for his invaluable guidance and continuous support throughout my project on Creating my own chatbot using Azure AI or any other Open Services.I extend my gratitude to Ms. Karthiga S, for guiding us into this internship and for her steadfast support and for creating an environment conducive to my learning and development by providing the necessary resources. Lastly, I am grateful to the institution for allowing me to undertake this project at the establishment. It has been a privilege to be part of the organization and to learn under their guidance.

Endorsement

I hereby declare that this internship report titled *"Internship Report on Microsoft Azure AI Intern – Edunet Foundations"* is my original work carried out during the internship period from 10th April 2025 to 15th May 2025.

This report has not been submitted for academic credit in any other institution.

Signature of the Student
Mohamed Azhmeer A.

Executive Summary

This report summarizes the work carried out during my internship at the Edunet Foundations, 10th April 2025 to 15th May 2025. Edunet Foundation is a social enterprise which was founded in 2015 and focuses on bridging the academia-industry divide, enhancing student employability, promoting innovation and creating an entrepreneurial ecosystem in India.

During the course of this internship, I was attached to the Microsoft Azure AI program under Edunet Foundation, which focuses on applying cutting-edge cloud-based artificial intelligence solutions to real-world applications in education and technology. The division emphasizes democratizing AI capabilities, fostering innovation, and enabling students to gain hands-on exposure to industry-relevant tools such as Azure OpenAI Services.

A major component of the internship centered around the design and development of a chatbot using Azure OpenAI Services. I explored prompt engineering, natural language understanding, and integration of conversational AI models into custom applications. Practical work included setting up the Azure environment, configuring API endpoints, managing authentication and resource deployment, and developing a chatbot that could respond intelligently to user queries.

Through this training, I developed competencies in cloud deployment, conversational AI development, debugging, and integration of AI services with application frontends. More importantly, I gained first-hand experience of how large language models and AI services can be harnessed to build scalable, intelligent systems, preparing me for future work in both research and industry settings.

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Company Overview

Name of the Organization: Edunet Foundation (in collaboration with Microsoft Azure AI)

Type: Non-Profit Organization & Training Partner

Sector of Activity: Education, Skill Development, and AI/Cloud Technology Enablement

Date of Creation: Established in 2015

Founder / Parent Organisation: Edunet Foundation, supported by Microsoft, IBM, and other industry partners

CEO / Director: Nagesh Singh (Founder & CEO, Edunet Foundation)

Headquarters Location: Bangalore, Karnataka, India

Number of Employees: Approx. 200+ including trainers, coordinators, and administrative staff, along with a nationwide network of academic collaborators

Strengths of the Organization:

- Strong industry–academia collaborations with global technology companies (Microsoft, IBM, AWS, etc.)
- Expertise in skill development programs in AI, Cloud Computing, Cybersecurity, and Emerging Technologies.
- Extensive reach through initiatives under *AI Skilling Programs*, *Microsoft Future Ready Talent*, and *IBM SkillsBuild*.
- Focus on bridging the gap between education and employability by providing hands-on projects and industry-standard tools.
- Proven track record of enabling thousands of students with future-ready skills aligned with *Digital India* and *Skill India* missions.

Internship

During the internship, I was engaged in tasks that familiarized me with Microsoft Azure AI services, the fundamentals of machine learning and deep learning, practical implementation of projects such as movie recommendation and sign language detection, and the end-to-end development of a chatbot using Azure OpenAI Services.

Setup of Edunet LMS and MS Learn

Context of the Task

The task involved completing structured learning modules provided by Microsoft Learn and Edunet LMS. The purpose was to gain foundational and advanced knowledge of Azure services, cloud computing, and AI/ML workflows. This served as the prerequisite for hands-on projects during the internship.

Tasks Assigned

1. Register and access MS Learn and Edunet LMS portals.
2. Complete foundational courses on Azure Cloud, Cognitive Services, and AI Services.
3. Attempt advanced learning paths related to Data Science, Machine Learning, and AI deployment.
4. Document key concepts and submit progress reports.
5. Submission of Completion Certificates in The LMS and Posts in LinkedIn

Solutions Proposed

i. Used the Microsoft Learn interactive sandbox for practical exercises.

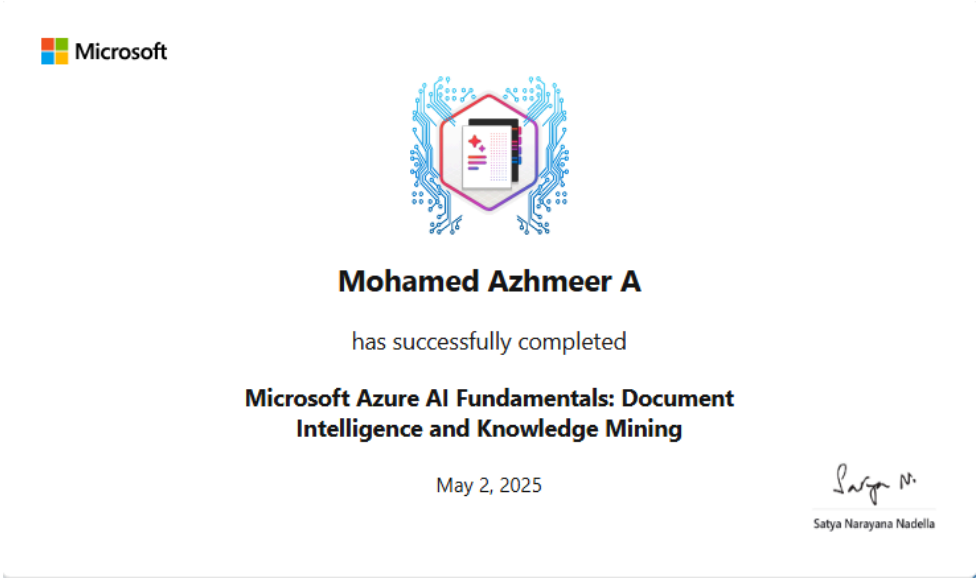
ii. tests to reinforce understanding.

iii. Created personal notes summarizing each module for quick revision during projects.

Observations

Output:

i. Successfully completed multiple modules and gained exposure to cloud service deployment.



Microsoft

Mohamed Azhmeer A

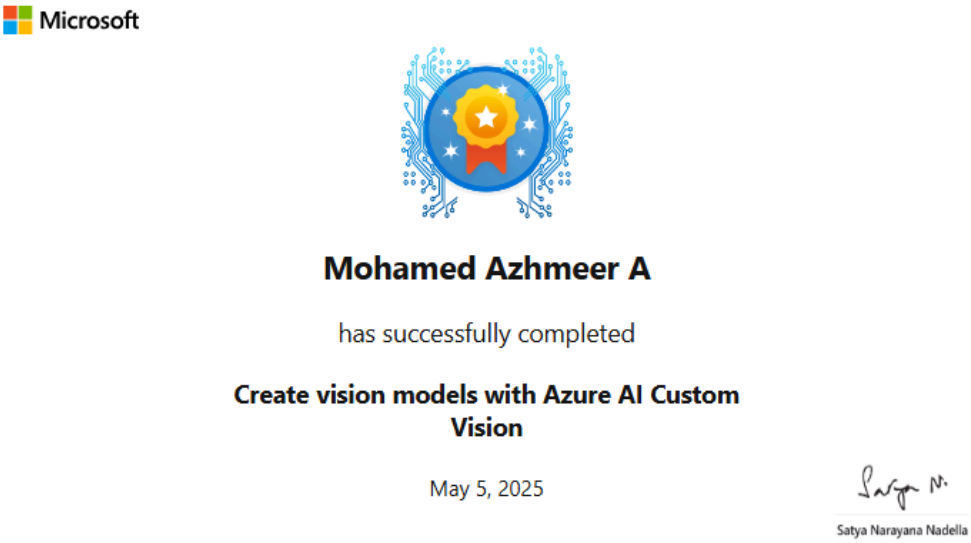
has successfully completed

Microsoft Azure AI Fundamentals: Document Intelligence and Knowledge Mining

May 2, 2025

Satya Narayana Nadella

ii. Understood core Azure AI capabilities like Language Services, Computer Vision, and Bot Framework.



Microsoft

Mohamed Azhmeer A

has successfully completed

Create vision models with Azure AI Custom Vision

May 5, 2025

Satya Narayana Nadella

Process :

i. Enrollment → Registered in MS Learn and Edunet LMS.

ii. Learning → Attended video-based sessions and hands-on sandbox labs.

iii. Assessment → Completed quizzes and module-end tests.

iv. Documentation → Compiled notes for future use in chatbot project.

Movie Recommendation System using Unsupervised Machine Learning

Context of the Project

The task focused on learning unsupervised machine learning, specifically clustering techniques, and applying them to a practical use case. A Movie Recommendation System was developed using datasets, where clustering algorithms grouped movies and provided recommendations based on similarities.

Tasks Assigned

i. Study clustering algorithms and their types.

ii. Set up the project environment using the GitHub repository provided by the mentor.

iii. Run the Movie Recommendation System and understand dataset preprocessing.

iv. Extend the project by experimenting with additional datasets and validating results.

Proposed Solutions

i. Used Python-based ML libraries (scikit-learn, pandas, numpy) for implementing clustering.

ii. Applied similarity measures to group movies into clusters.

iii. Visualized clusters and validated recommendations against sample queries.

Observation

i. Successfully executed the project and verified outputs.

ii. Understood the importance of preprocessing (removing nulls, normalization).

iii. Learned how clustering improves personalization in recommendation systems.

Process

i. Setup → Cloned the GitHub repository and configured Python environment.

ii. Data Preprocessing → Loaded dataset, handled missing values, normalized features.

iii. Model Training → Implemented clustering algorithm (K-means).

iv. Validation → Tested system with multiple inputs and verified outputs.

Output :

```
import pandas as pd
import numpy as np
import ast  # for parsing genre strings
from sklearn.preprocessing import MultiLabelBinarizer, MinMaxScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import seaborn as sns
import matplotlib.pyplot as plt
```

```
[17]: # Load the movie metadata CSV file
df = pd.read_csv("movies_metadata.csv", low_memory=False)
```

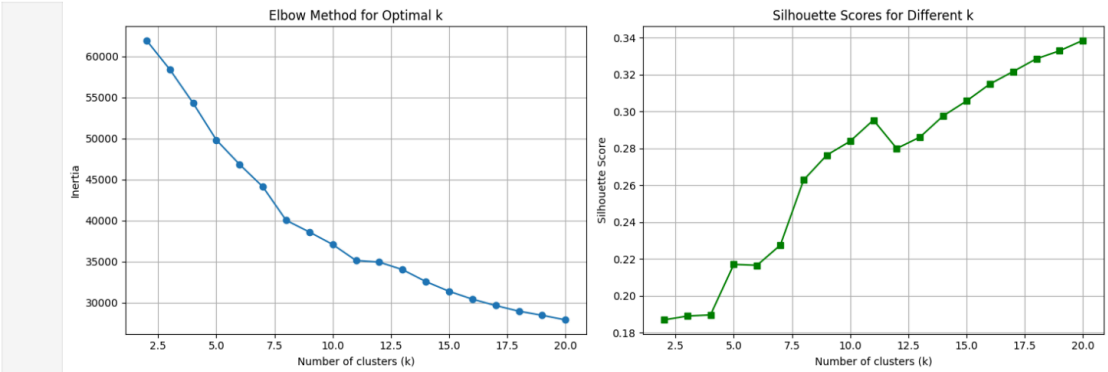
```
[18]: #checking all the columns from datasets
df.columns
```

```
[18]: Index(['adult', 'belongs_to_collection', 'budget', 'genres', 'homepage', 'id',
       'imdb_id', 'original_language', 'original_title', 'overview',
       'popularity', 'poster_path', 'production_companies',
       'production_countries', 'release_date', 'revenue', 'runtime',
       'spoken_languages', 'status', 'tagline', 'title', 'video',
       'vote_average', 'vote_count'],
      dtype='object')
```

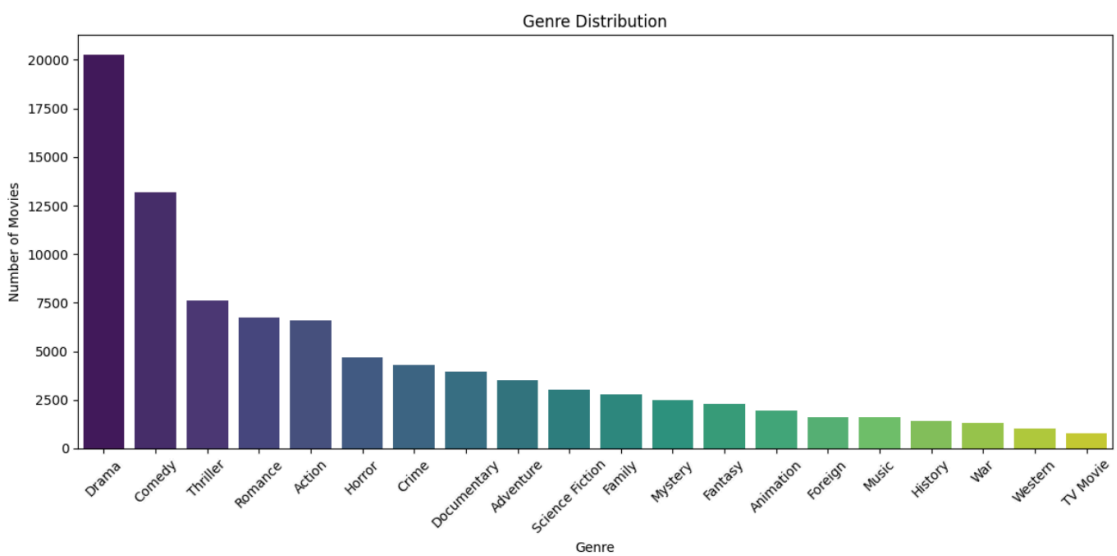
```
[19]: # We only need title, genres, vote_average, and popularity for this project
movies = df[['title', 'genres', 'vote_average', 'popularity']].copy()
```

```
[20]: # Drop rows with missing values in important columns
movies = movies.dropna(subset=['genres', 'vote_average', 'popularity'])
```

```
[24]: #Extract genre names from JSON-like string
def parse_genres(genre_str):
    """Converts a string of genre dictionaries into a list of genre names."""
```




```
[27]: genre_counts = movies['genre_list'].explode().value_counts().sort_values(ascending=False)

plt.figure(figsize=(12, 6))
sns.barplot(x=genre_counts.index, y=genre_counts.values, palette='viridis')
plt.title('Genre Distribution')
plt.xticks(rotation=45)
plt.ylabel('Number of Movies')
plt.xlabel('Genre')
plt.tight_layout()
plt.show()
```



Sign Language Detection Project using Deep Learning

Context of the Project

The task was aimed at understanding the fundamentals of deep learning and applying them to a real-world application — detecting and classifying sign language gestures. This project helped build skills in neural networks, image classification, and human-computer interaction.

Tasks Assigned

i. Study the basics of deep learning and neural networks.

ii. Set up and run the GitHub project for Sign Language Detection.

iii. Train the model on the provided dataset and observe accuracy results.

iv. Demonstrate outputs by classifying real-time or test gesture images.

Proposed Solutions

i. Used TensorFlow/Keras frameworks for training CNN-based models.

ii. Preprocessed input images (resizing, grayscale, normalization).

iii. Augmented dataset for improved accuracy and robustness.

Observation

i. Successfully ran the project and generated accurate predictions.

ii. Understood challenges like dataset imbalance and overfitting.

iii. Recognized the scope of deep learning in accessibility and assistive technologies.

Process:

i. Setup → Cloned repository and installed dependencies.

ii. Data Preparation → Preprocessed dataset images.

iii. Training → Executed CNN model and tuned hyperparameters.

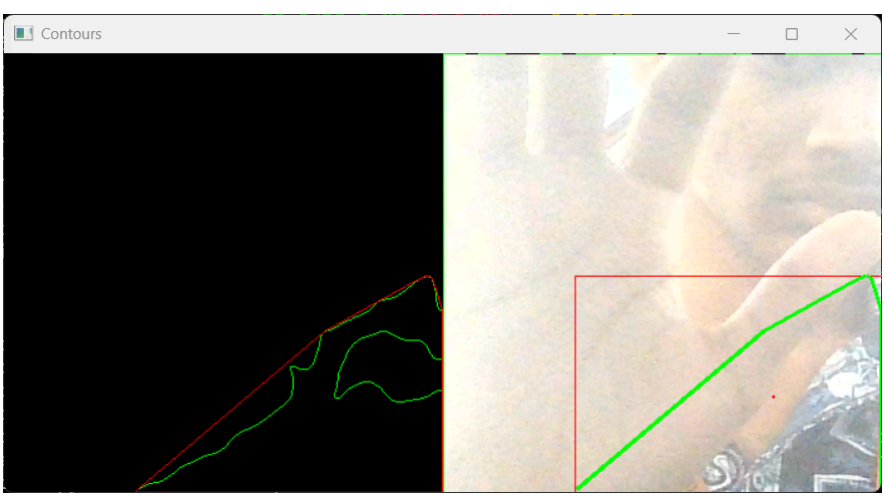
iv. Validation → Tested model predictions on sample gestures.

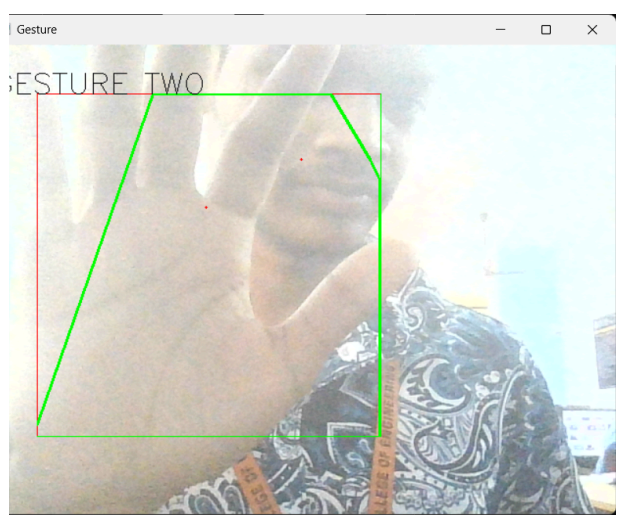
Output:

HandGestureRecognitionOpenCV.py

Code > HandGestureRecognitionOpenCV.py

```python
import cv2
import numpy as np
import math
cap = cv2.VideoCapture(0)
while(True):

    ret, img = cap.read()
    if not ret:
        print("Failed to capture frame")
        continue
    if img is None:
        print("No frame captured")
        continue

    cv2.rectangle(img,(400,400),(50,50),(0,255,0),0)
    crop_img = img[50:400, 50:400]
    grey = cv2.cvtColor(crop_img, cv2.COLOR_BGR2GRAY)
    value = (35, 35)
    blurred = cv2.GaussianBlur(grey, value, 0)
    _, thresh1 = cv2.threshold(blurred, 127, 255,
                               cv2.THRESH_BINARY_INV+cv2.THRESH_OTSU)
    cv2.imshow('Thresholded', thresh1)

    contours = cv2.findContours(thresh1.copy(), \
           cv2.RETR_TREE, cv2.CHAIN_APPROX_NONE)[-2]

    cnt = max(contours, key = lambda x: cv2.contourArea(x))

    x,y,w,h = cv2.boundingRect(cnt)
    cv2.rectangle(crop_img,(x,y),(x+w,y+h),(0,0,255),0)
    hull = cv2.convexHull(cnt)
    drawing = np.zeros(crop_img.shape,np.uint8)
    cv2.drawContours(drawing,[cnt],0,(0,255,0),0)
    cv2.drawContours(drawing,[hull],0,(0,0,255),0)
    hull = cv2.convexHull(cnt,returnPoints = False)
    defects = cv2.convexityDefects(cnt,hull)
    count_defects = 0
```





Creation of Chatbot using AI Services

Context

The final and most significant task involved designing and developing a chatbot called "**Remify**" using Azure OpenAI Services. The goal was to integrate a conversational AI system capable of responding intelligently to user queries.

Tasks Assigned

i. Set up Azure environment and create an OpenAI resource.

ii. Configure API keys, authentication, and endpoints.

iii. Develop chatbot backend using Node.js and integrate with Azure services.

iv. Deploy chatbot and test interaction flow.

Solutions Proposed

i. Used Azure OpenAI Service for accessing GPT-based models.

ii. Integrated backend with API endpoints to handle chat requests.

iii. Added custom features such as memory, context retention, and error handling.

iv. Deployed project on GitHub for version control and collaboration.

Observation

i. Successfully built a functional chatbot capable of handling user inputs.

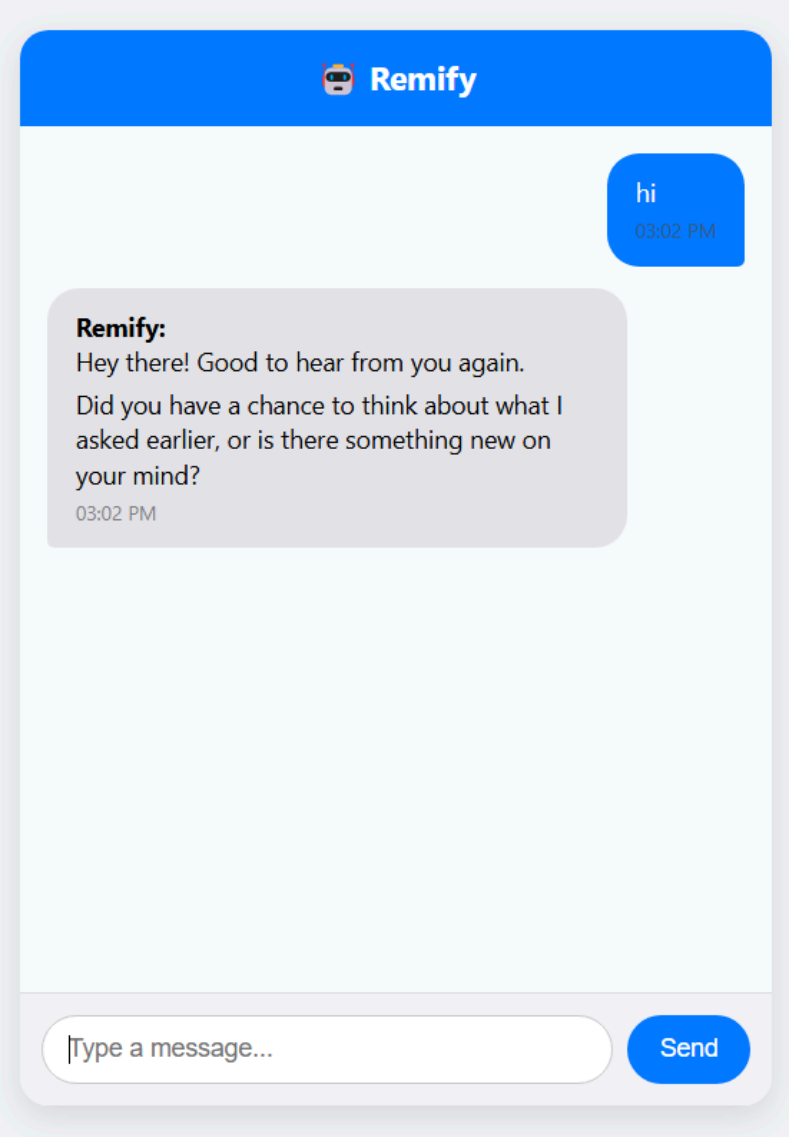
ii. Learned prompt engineering and optimization techniques for better responses.

iii. Understood deployment pipelines and version control using GitHub.

Process:

i. Environment Setup → Configured Azure AI services and API credentials.
ii. Backend Development → Implemented chatbot logic using Node.js.
iii. Testing → Ran multiple test cases to verify responses.
iv. Deployment → Hosted project and documented installation steps on GitHub.

Output:



Analysis of my experience

During the setup task, several important observations emerged. Technically, the process required configuring Azure resources, managing API keys, and integrating development environments with OpenAI endpoints, where correct authentication and deployment steps were critical. On the human side, collaboration with mentors and guided sessions were essential in overcoming challenges. A clear gap between theory and practice also surfaced: while academic resources focus on AI concepts like machine learning models and conversational design, the real-world setup demanded handling environment configurations, version compatibility, API response debugging, and resource management on the Azure portal. The main difficulties involved authentication errors, dependency issues, limited documentation for specific use cases, and interpreting error logs during testing. Despite this, successfully establishing a working Azure AI environment was highly rewarding, enhancing practical skills, problem-solving abilities, and bridging the gap between theoretical AI concepts and their real-world implementation.

Conclusion:

Working on this internship gave me a hands-on understanding of how AI services can be applied to solve real-world problems, particularly through the development of a chatbot using Microsoft Azure OpenAI Services. Concepts like natural language processing, machine learning models, and cloud-based deployment moved from being abstract ideas to practical tools that I could implement and experiment with.Beyond the technical knowledge, the experience also strengthened my ability to manage time, take initiative in projects, and work independently while still seeking guidance when needed. I learned that true mastery comes not only from completing guided tasks but also from consistently experimenting, debugging, and refining solutions.This internship has sparked my curiosity to explore advanced AI frameworks, cloud integration techniques, and conversational AI systems. The skills and insights I gained will serve as a strong foundation for my future career in IT, artificial intelligence, and software development.

References:

MS Learn : https://learn.microsoft.com/en-us

Edunet Foundations :https://edunetfoundation.org

Git Hub Repositories:

- Movie Recommendation System : https://github.com/therohitshelar97/MovieRecommendationUsingUnsupervisedLearning
- Sign Language Detection : https://github.com/therohitshelar97/SignLanguageDetectionDeepLearningProject

Remify Chatbot: https://remify.onrender.com